

3-h Water Jug in Prolog

PROGRAM:

```
% State represented as state(Jug1, Jug2)

move(state(X, Y), fill1, state(4, Y)) :- X < 4.
move(state(X, Y), fill2, state(X, 3)) :- Y < 3.
move(state(X, Y), empty1, state(0, Y)) :- X > 0.
move(state(X, Y), empty2, state(X, 0)) :- Y > 0.
move(state(X, Y), pour12, state(X1, Y1)) :-
    X > 0, Y < 3,
    Transfer is min(X, 3 - Y),
    X1 is X - Transfer,
    Y1 is Y + Transfer.
move(state(X, Y), pour21, state(X1, Y1)) :-
    Y > 0, X < 4,
    Transfer is min(Y, 4 - X),
    X1 is X + Transfer,
    Y1 is Y - Transfer.

solve(State, Goal, Path) :- bfs([[State]], Goal, Path).

bfs([[Goal|Rest]|_], Goal, [Goal|Rest]).
bfs([Path|Queue], Goal, Solution) :-
    extend(Path, NewPaths),
    append(Queue, NewPaths, Queue1),
    bfs(Queue1, Goal, Solution).

extend([State|Rest], NewPaths) :-
    findall([NewState,State|Rest],
        move(State, _, NewState),
        NewPaths).

run_jug :-
    writeln('Initial state is (0,0). Goal is (2,0).'),
    solve(state(0,0), state(2,0), Path),
    writeln('Solution Path:'),
    writeln(Path).
```

OUTPUT:

A screenshot of the SWI-Prolog console window. The window has a title bar with a red icon and the text "SWI-Prolog console". Below the title bar is a menu bar with "File", "Settings", "Tools", "Debug", and "Help". The main area is a black terminal with white text. It shows the welcome message, a license notice, and help instructions. Then, it shows the execution of a Prolog program that consults a file and runs a game, displaying the solution path and ending with a red triangle cursor.

```
SWI-Prolog console
File Settings Tools Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

30 ?- consult(['c:/Users/kames/OneDrive/Documents/jugg.pl']).
true.

31 ?- run_jug.
Initial state is (0,0). Goal is (2,0).
Solution Path:
[state(2,0),state(0,2),state(4,2),state(3,3),state(3,0),state(0,3),state(0,0)]
true ▲
```